

KROPPEFJALL-GRANAN, SwedenWMO: **025400**Lat: **58.6065N**Lon: **12.1973E**Elev: **174**StdP: **99.25**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.2	-11.0	-17.6	0.8	-10.0	-14.3	1.1	-10.0	7.0	4.7	6.3	4.0	1.3	60	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	25.6	17.8	23.8	17.0	22.0	16.1	19.0	23.4	18.1	22.3	17.1	20.6	1.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	12.8	20.2	16.5	12.0	19.3	15.6	11.3	18.4	54.6	23.5	51.7	22.3	48.6	20.6	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	4.7	DB	-15.0	28.0	3.5	1.4	-17.6	29.0	-19.6	29.9	-21.6	30.7	-24.2	31.7	(4)
(5)			WB	-15.3	20.6	3.5	1.1	-17.8	21.4	-19.8	22.1	-21.8	22.7	-24.3	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-1.9	-1.9	1.0	5.8	10.5	14.2	16.5	15.4	11.7	6.9	2.7	0.0		
	DBStd	7.40	4.30	4.23	3.87	3.26	3.59	2.85	2.71	2.31	2.76	3.33	3.70	4.64		
	HDD10.0	1797	369	333	279	132	37	2	0	0	15	104	218	309		
	HDD18.3	4234	627	567	537	376	245	128	72	94	200	353	468	567		
	CDD10.0	629	0	0	0	6	51	128	202	168	65	9	0	0		
	CDD18.3	24	0	0	0	0	1	4	15	4	1	0	0	0		
	CDH23.3	198	0	0	0	0	6	48	121	21	1	0	0	0		
	CDH26.7	19	0	0	0	0	0	4	15	1	0	0	0	0		
(14) Wind	WSAvg	2.1	2.3	2.2	2.4	2.2	2.0	2.0	1.9	1.7	1.9	1.9	2.2	2.4		
(15) Precipitation	PrecAvg	951	78	62	50	56	63	84	92	87	86	103	93	89		
	PrecMax	1280	160	134	116	128	85	137	202	179	194	227	175	188		
	PrecMin	679	12	18	19	9	23	45	21	25	26	32	26	16		
	PrecStd	165	43	33	26	38	20	30	44	40	46	52	42	49		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	8.3	13.3	18.9	24.2	27.1	28.5	25.5	22.2	15.8	11.5	9.3		
		MCWB	5.5	4.8	8.0	10.7	14.7	17.1	18.7	18.2	16.9	13.0	10.7	8.7		
	2%	DB	5.7	5.7	10.0	16.3	21.8	24.6	26.2	23.6	19.1	13.6	10.1	8.1		
		MCWB	4.6	4.2	5.8	9.9	13.7	16.4	18.3	17.8	15.0	12.0	9.3	7.5		
	5%	DB	4.8	4.6	7.5	13.7	19.4	22.3	24.6	21.7	17.2	12.6	9.1	7.1		
		MCWB	3.5	3.6	4.4	8.2	13.0	15.3	17.9	16.7	14.1	11.4	8.5	6.2		
	10%	DB	3.9	3.6	5.9	11.4	16.8	20.1	22.4	19.8	15.9	11.6	7.9	6.0		
		MCWB	2.8	2.8	3.8	7.0	11.8	14.4	17.0	15.7	13.4	10.5	7.3	5.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.7	8.2	12.0	16.8	18.8	20.7	20.0	17.7	13.7	11.2	8.9		
		MCDB	6.7	6.7	12.2	17.6	21.8	23.4	25.0	22.9	20.7	14.6	11.3	9.0		
	2%	WB	4.9	4.6	6.5	10.3	15.1	17.3	19.5	18.5	15.8	12.6	9.6	7.7		
		MCDB	5.6	5.2	9.0	14.9	18.8	22.8	24.2	22.0	17.9	13.3	9.8	8.0		
	5%	WB	3.9	3.7	5.1	9.0	13.7	16.2	18.5	17.4	14.8	11.6	8.5	6.3		
		MCDB	4.5	4.4	7.0	12.8	17.5	20.9	23.2	20.5	16.7	12.4	8.9	6.8		
	10%	WB	3.0	2.7	4.1	7.7	12.4	15.0	17.7	16.5	13.8	10.8	7.4	5.1		
		MCDB	3.7	3.6	5.4	10.5	15.8	18.9	21.6	19.0	15.4	11.5	7.9	5.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.4	6.2	8.3	8.9	8.8	8.5	7.7	7.1	5.4	4.0	3.9		
		MCDBR	4.8	5.0	8.7	12.0	11.9	12.1	12.1	10.5	8.8	6.0	4.9	4.4		
	5% WB	MCWBR	4.8	4.0	5.6	7.0	6.4	6.3	5.9	5.6	5.2	4.5	4.8	4.5		
		MCWBR	4.9	4.6	7.8	10.5	10.1	10.7	10.6	9.0	8.0	5.5	4.7	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.312	0.332	0.369	0.365	0.354	0.375	0.371	0.341	0.330	0.298	0.260		
	taud		2.235	2.317	2.375	2.330	2.408	2.460	2.419	2.427	2.499	2.438	2.299	2.162		
	Ebn at Noon		551	715	813	836	868	883	852	828	801	691	523	436		
	Edh at Noon		45	69	86	106	105	101	103	96	76	61	43	34		
(44) All-Sky Solar Radiation	RadAvg		0.33	0.90	2.42	3.92	5.21	5.89	5.29	4.16	2.80	1.26	0.44	0.23		
	RadStd		0.06	0.16	0.23	0.50	0.45	0.32	0.62	0.34	0.32	0.20	0.09	0.02		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page